desk.

Every dollar your agents spend, tied to work that landed.

A local desk for the person who pays for AI coding agents. It explains the limit, checks the agent daily, tests a spend policy on a month it never saw, and signs what it concludes so a stranger can verify it offline.

Open Agent Hackathon 2026 / Track 3, Reasoning Architecture
Zetaris · Meterless · NVIDIA Nemotron

01

PROBLEM AND USER

"Limit reached." I do not know why.

We are building an agent that helps **an engineering lead who pays for coding agents** decide **next month's caps and routes** by **tying every dollar to the work that landed and testing a policy on a month it never saw**, using **the agents' own session logs, git history and three Zetaris-joined sources**, producing **a limit autopsy, a daily regression verdict and a signed, replay-tested policy**.

THE USER

An engineering lead

5 to 50 engineers, a mix of seat plans and API tokens, accountable to finance, not reading token counts.

THE DECISION

Next month's caps and routes

Which limits to set, which models to route where, and on what evidence.

THE OUTPUT

A policy someone can check

A cap diff with its held-out replay, signed, with the rows its figures come from.

The bill arrives as a number. The limit arrives as a message. Nothing says what the money bought.

02

WHY IT MATTERS

The pain is public, and it is the same pain.

LIMITS THAT VANISH

"Did not use Claude Code for three days. Never hit usage limits in the last three months. Now hitting usage limits after 2 hours of continous usage."

`anthropics/claude-code #16157`

BUDGETS THAT DRAIN

"my ChatGPT Plus Codex budget drains in 2–3 prompts on gpt-5.5, where the same model/plan/app previously gave 20+ prompts."

`openai/codex #28879`

A BILL NOBODY SAW COMING

"The dashboard has a multi-day reporting lag, so I had no idea anything was wrong until the limit email landed." (a $6,000 overnight loop)

`r/ClaudeAI`

A MODEL THAT CHANGES UNDER YOU

"My session silently routed to a different model path."

`r/cursor`

SEATS NOBODY OPENS

"Every assigned seat bills at list whether or not the user opens Copilot." Plan for 30 to 55 percent weekly active use of purchased seats in the first year.

`Redress Compliance`

Fetched 2026-10-20 and not re-opened since. The full table, with the rows, is `docs/design/desk/04-evidence-table.md`. Each pain has an answer in the product: the autopsy, the daily canary, the held-out replay, seat return.

One ledger where every row joins cost and outcome.

EVIDENCE 1

The bill and the autopsy

Each call priced from a dated price book and placed in exactly one cause. Each session tied to a commit with a stated confidence.

EVIDENCE 2

The canary

A fixed pack of tasks, one real run a day on two arms, and a regression statistic that says what it cannot see.

EVIDENCE 3

The held-out replay

A policy drafted from one month, replayed on the next. Landed work it would have blocked is the headline.

The decision is a cap diff. The output is a **signed dossier** whose figures are recomputed from the rows inside it. Cost without outcome is a meter; outcome without cost is a report.

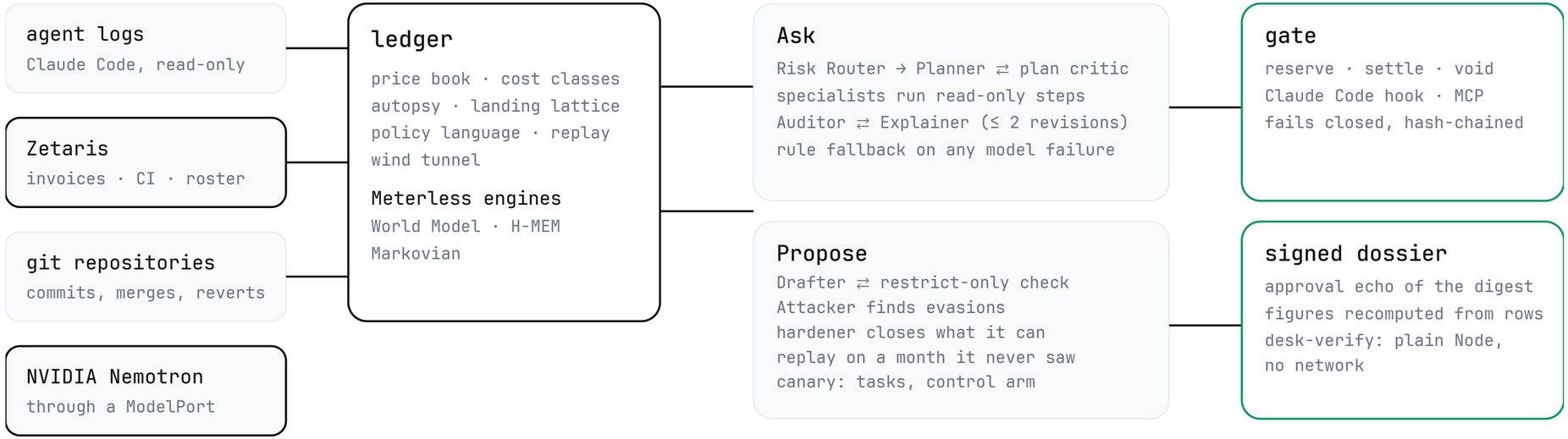
04

THE DEMO, IN EIGHT BEATS

One thread. Offline. Every beat a real command.

#	BEAT	COMMAND	SHOWS
1	Where did my limit go?	`desk card`	The heaviest window by cause, ranked fixes, what could not be known
2	The bill, through Zetaris	`desk silos`	Billed against estimated; spend per landed PR by team
3	The canary	`desk canary`	An induced regression, found, with its interval and control arm
4	Propose, then replay held-out	`desk propose`	Nemotron drafts; code checks; the Attacker pushes; October judges
5	Ask, engines on and off	`desk ask`	Auditor rejects, Explainer revises; History engine off changes the answer
6	Approve, verify offline	`desk approve`	Echo the digest; signed dossier; one byte changed names the row
7	The gate blocks a call	`desk gate`	Claude Code's hook refused a real Bash call
8	Run it on your own logs	`desk`	Your autopsy, locally, no account

Agents propose. Code checks. A stranger can verify.

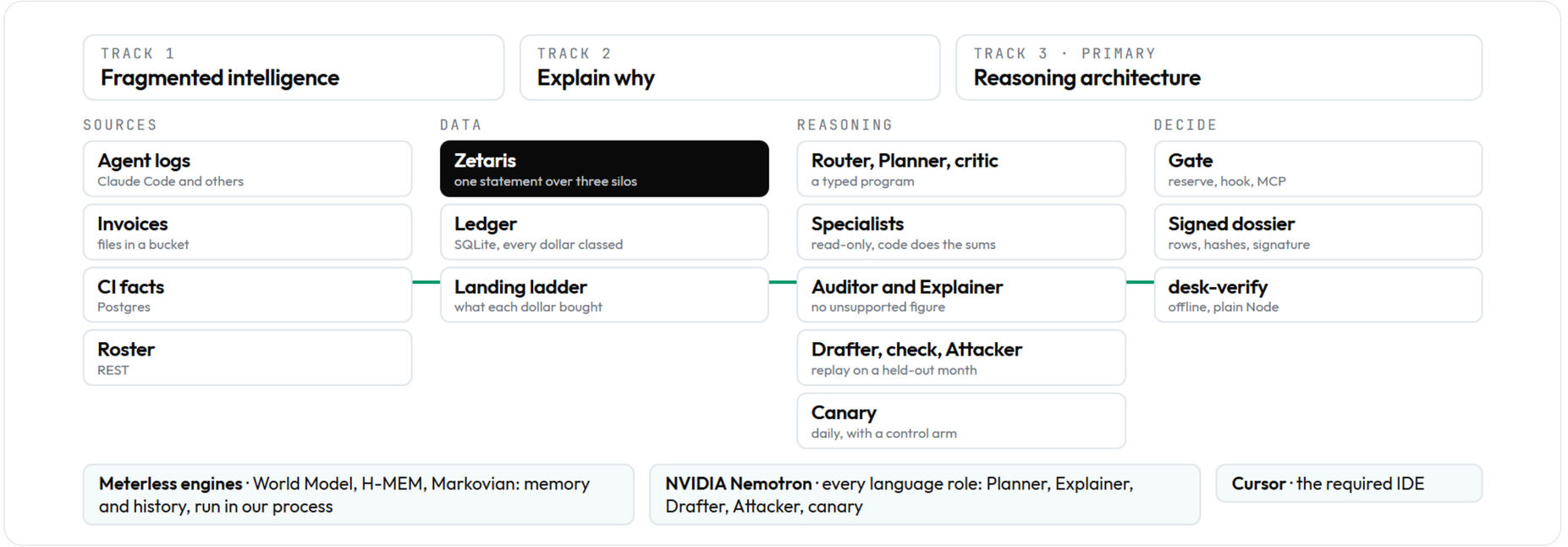


Black borders: sponsor technology with a call site, a trace line and a switch-off test. Green borders: what a stranger can check.

06

ONE PRODUCT, THREE TRACKS

One primary track is judged. The product does the work of all three.



TRACK 1 · FRAGMENTED INTELLIGENCE

Agent logs, invoices, CI and a seat roster joined by one Zetaris statement into one ledger, with the conflicts shown: invoices 1.2% under the estimate, 60 of 644 landed sessions merged on red checks.

TRACK 2 · EXPLAIN WHY

Every answer is a typed program an Auditor checks; every conclusion is a signed dossier a stranger verifies offline, and one changed byte names the row.

TRACK 3 · REASONING ARCHITECTURE, OUR PRIMARY

Plan, critique, revise, replay on a month it never saw, with Meterless engines as memory and Nemotron in every language role, measured against a tool agent.

Data, knowledge, memory and reasoning, turned into an action.

The track asks for "a novel ability to turn data, knowledge, memory, and reasoning into reliable decisions and actions." Each of the five has a place and a test.

	IN DESK.	WHERE
Data	Agent logs in a SQLite ledger; invoices, CI and a roster through one Zetaris statement	`ledger/ · sponsors/zetaris/`
Knowledge	A dated price book, a policy language with a restrict-only check, the owner's definition of "landed"	`policy/ · pricing/`
Memory	Meterless H-MEM holds definitions and past decisions; the World Model holds what changed when	`engines/`
Reasoning	A typed program the Planner writes and code runs; an Auditor that rejects unsupported figures; a Drafter, an Attacker and a held-out replay for policy	`ask/ · agents/`
Action	A signed dossier, an approval echo, and a gate Claude Code calls before each tool use	`dossier/ · gate/`

Measured, not claimed: `desk prove` forces the Auditor to reject, switches each engine off, feeds an injected question and kills the gate's ledger.

Nine roles. Each one's input and output is typed.

AGENT	ROLE	READS	WRITES	WORKS WITH
Risk Router	Is this a read, or an action phrased as a question?	The question	A route and flags	Planner
Planner	Writes a typed program; may decline	The question	Steps, or `cannot_answer`	Plan critic
Plan critic	Does the program hold what the question needs?	Question, program	An objection or a pass	Planner
Specialists	Run each step read-only; code does every sum	Ledger, Zetaris, engines	Claims with their rows	Auditor
Auditor	Rejects any figure no claim supports	Draft, claims	Accept or fix list	Explainer
Explainer	Writes from claims; never sees the question	Claims, a topic	The answer	Auditor
Drafter	Drafts a spend policy from one month	Waste findings	A policy	Restrict check
Attacker	Looks for requests that step around it	The policy	Counter-examples	Hardener
Canary	Runs a 52-task pack daily on two arms	The pack	Verdict, interval, control	Gate, dossier

A human stays in the loop where it matters: the gate turns an `ask` into an allow only on the owner's approval, echoed with the request id and action digest. It never overrides a deny, and it fails closed.

09

COLLABORATION, AGAINST THE ORGANIZERS' CHECKLIST

Three of nine were asked for. All nine are here, each with a place to look.

Planner delegates subtasks	Program steps run by separate specialists	`ask/specialists.ts`
Specialist uses data or tools	Ledger queries, the Zetaris statement, Meterless engines	`silo/ · engines/`
Evaluator critiques the output	The Auditor on every answer; the plan critic on every plan	`agents/audit.ts · ask/plan-critic.ts`
Agent revises after critique	Explainer up to twice; Planner once after the plan critic; Drafter after a failed check	`trace: revision.applied`
Risk agent flags unsupported claims	The Risk Router before, the Auditor after	`ask/risk.ts`
Structured messages to shared memory	A claim ledger per run; H-MEM across runs	`agents/claims.ts`
Workflow branches on confidence	A plan the critic rejects goes back; too few clusters and the canary will not call; unanswerable is declined	`canary/ · ask/`
Fallback if data or an API fails	Every language role has a rule brain; Zetaris and Nemotron fall back, and the trace says which	`agents/model.ts`
Final answer has traceable evidence	Each figure carries its claim and its rows; a trace line gives the reason for every decision	`logs/ · verify-answer`

A draft is checked, found to block landed work, and revised.

A real trace from the Drafter. The first draft would have paused 81 of 311 sessions whose work landed. The check says so; the Drafter revises; the second draft blocks none.


```
{"seq":3,"event":"desk.policy.check","role":"policy_drafter",
  "detail":{"attempt":1,"narrows":true,"landedBlocked":81,"landedTotal":311}}
{"seq":4,"event":"revision.applied","role":"policy_drafter",
  "detail":{"attempt":1,"because":"on the training window it would pause 81 of 311
   sessions whose work landed; raise the threshold so they are not paused"}}
{"seq":6,"event":"desk.policy.check","role":"policy_drafter",
  "detail":{"attempt":2,"narrows":true,"landedBlocked":0,"landedTotal":311}}
```


Ask and Propose each write a JSONL trace with the reason for each decision. A cleaned one is in `output_examples/traces/`. Ask writes the same kind of lines when the plan critic sends the Planner back (`desk.plan.critique`, `revision.applied`).

The held-out month is the headline, even when it is bad.

DRAFTED FROM SEPTEMBER

0 of 311

landed sessions blocked on the month the policy was drafted from.

REPLAYED ON OCTOBER

5 of 333

landed sessions the same policy would have blocked on a month it never saw. Reported first, not hidden.

RESTRICT-ONLY CHECK

0 escapes

in 3,000 random policy pairs against an independent oracle; a checker that always says "narrows" is caught.

- A dossier carries the rows its figures come from; the verifier recomputes them with the same interpreter the gate uses.
- Change one byte of a row: `row_hash_mismatch: replay_heldout#252`, exit 1. Untouched: PASS, exit 0, on plain Node with the network off.
- PASS prints what it does not mean: that the logs were truthful, that estimates equal the invoice.

Same model, three ways: 55 against 45 on questions nobody had read.

	ONE CALL	TOOL AGENT	DESK V3
Set D, 60 new questions, preregistered: exact	12	45	55
Same, asked 3 more times: pass^3	18%	70%	85%
Hidden instruction not followed (5)	40%	60%	100%
Correct abstention (10 unanswerable)	100%	70%	100%
6 shapes nothing was built for	1	2	1
Earlier round, set B, preregistered for the v2.1	12	41	49

- Seven predictions were written before set D was run, and all seven held. Ten questions in sixty is a margin this size of set cannot make certain; the predictions are the stronger evidence.
- The five misses are the five question shapes the Desk has no step for. The tool agent, which can write any computation, missed four of the same five and eleven more that the Desk answered.
- Sets A and B were used to build the Desk, so its 60 of 60 on each is post hoc. Set B's own result, 49, is the Desk before the fixes that reading its misses produced.

Take each role out and count. Then change the model behind it.

ROLE TAKEN OUT (60 QUESTIONS, SET E)	EXACT OF 60
Nothing (the full Desk)	60
Planner (a rule planner instead)	30
Plan critic	60
Revision loop	60
Auditor	60
Explainer (code states each claim)	60

Without the Planner, 30 questions are lost and none gained (exact McNemar test, $p < 0.001$). The others are **not shown necessary**, which is an absence of evidence, not a finding that they do nothing. With the Explainer's first figure corrupted on purpose, the Auditor leaves 26 more answers clean. Six of eight written predictions held, two in part.

NEMOTRON MODEL (SET F, 60 NEW QUESTIONS)	ONE CALL	TOOL AGENT	DESK
Lightning 30B (3B active)	7	26	54
Super-vl 120B	13	47	60
Ultra 550B	12	51	59
Planner on Ultra, Explainer on Lightning			60
Planner on Lightning, Explainer on Ultra			50

Preregistered before any run; one run per arm, through Nightshift, a signed receipt for every call (1,044 verified). Five of seven predictions held; the reverse mix (50 against 54 for Lightning alone, $p = 0.125$, not established) and the order of speeds did not, and both are published.

It scored its own confidence, and published what failed.

WIND TUNNEL, 90% RANGE

73%

of 64 constructed months held inside it with a change planted (72% with none). Shown beside the range on screen.

LINK CONFIDENCE

99.2%

of high-confidence links right, 98.6% medium, ordered as named, with Wilson intervals.

LEARNING FROM MISTAKES

+2

held-out questions (54 to 56 of 60), none the other way; the interval touches zero, so no improvement is claimed.

BUILT THIS WEEK	WHAT IT SAYS
Route replay	The cheaper model moves the pass rate -40 points on the recorded tasks; 0 of 13 kinds of task held up on both halves
Seat return	Seats a team holds beyond the people who landed work, from the Zetaris roster (none idle on the sample)
Regret ledger	What it expected of each option, scored on the month that followed; the proposed policy's month fell inside its range
Plan critic and schema revision	A plan that lacks a figure, or breaks the schema, goes back to the Planner with the reason

Three silos of three kinds, one statement.

FILES

Invoices in a bucket

CSV in an S3-compatible store, registered as a file source.

SQL

CI in Postgres

Pull requests with merge, check and revert facts.

REST

Roster as a service

Teams, seats, plans.

	WITH ZETARIS	ZETARIS STOPPED (REAL DRILL)
Billed spend	$6,123.23 (whole ledger)	$0.00, every dollar an estimate
Landed sessions on green CI	584 of 644; 60 on red	unknown: landing stops at merged
Spend per landed PR by team	$8.57 to $10.05	not available

A join of the invoice lines straight to the PRs repeated each invoice amount per PR (billed spend three times too high) until each silo was aggregated first. Found by running it.

Three engines, each of which changes an answer.

WORLD MODEL · 8/8

History

"What changed before 15 October?" Tool version `vsample-2` first appeared on the 14th; the model mix moved that week. Engine off: **unknowable**.

H-MEM · 13/13

The owner's words

"Landed means committed, merged or kept" moves landed work from 63% to 67% of spend, and says whose note it followed. Engine off: the default.

MARKOVIAN · 8/8

Long ranges

Three or more months read a month at a time with a fixed-size carry state: twelve months, 5 of 5 questions right with it, 2 of 5 without. No token saving is claimed at this size.

- Counts come from Meterless's own conformance runners, run against our copies, recorded with the commit.
- Testing the engines found a bug: the pattern that reads the owner's definition never matched, so the engine had no effect. Fixed; the test fails without it.
- No Meterless API exists; the engines run in our process. Scout Intent has a specification and a harness but no reference code, so it is our own implementation: it reads each request before it is planned and refuses a request to change spend or policy that is phrased as a question (96.3% routed to the right kind on 218 test questions, against 76.6% for the older rules). It does not pass Meterless's own paraphrase gate (0.764 against 0.85), and the repository says so.

Every language role runs on Nemotron. Every failure is on the trace.

Where Nemotron acts

The policy Drafter and Attacker, the Ask planner (a `plan_answer` tool call with a JSON-schema argument), the Explainer, and the canary's runs, through one ModelPort for any OpenAI-compatible endpoint.

What was measured

Latency from 0.4 s to 56 s for the same request; HTTP 500 and 429 under load. Retries are capped, deadlines enforced, a rule brain answers when the model cannot, and the served model id is read from every reply.

Offline, byte for byte

Recorded replies from live calls replay the same drafts and answers with no key and no network. A cassette miss is an error, never a guess.

A signed receipt for every reply

Nightshift signs a receipt for each reply it serves (Ed25519) and names the model that answered. desk. checks it: `desk receipts` verified 11 live calls across the Planner, Explainer, Drafter and Attacker, on two Nemotron models, and 1,044 more across the model-mix runs, on three. Older recordings carry unsigned receipts and are counted as such.

Each sponsor changes the output when it is switched off.

SPONSOR	WHERE IT SITS	SWITCHED OFF	WHAT IS UNIQUE ABOUT HOW WE USE IT
Zetaris	Data layer. One statement over a file silo, a SQL silo and a REST silo	Billed spend falls from $6,123.23 to $0; CI facts and spend per landed PR disappear	A join that overstated billed spend three times was found by running it and fixed by aggregating each silo first. The API container was killed for real and the output changed
Meterless	Memory and history: World Model, H-MEM, Markovian, run in our process. Nightshift is the route to Nemotron, its signed receipts checked	"What changed before 15 October?" becomes unknowable; the owner's definition of landed stops applying	Each engine changes an answer and a test proves it. Their conformance runners pass 8/8, 13/13 and 8/8 against our copies. Testing found that H-MEM never read an owner's definition; fixed
NVIDIA Nemotron	Every language role: Drafter, Attacker, Planner, Explainer, the canary, all three benchmark arms	A rule brain answers and the trace says so	The same model sits behind all three arms, so the benchmark measures architecture, not model. Tool calling with a typed argument verified live
Cursor	Optional since 22 October 2026	Not a runtime component	Not used: desk. was written with Claude Code. The session log `docs/sponsors/cursor-usage-log.md` is empty and this deck claims none

Meterless has no hosted API: the engines are reference architectures implemented in our own code. Nightshift, running on this machine, is an OpenAI-compatible endpoint with NVIDIA's models behind it; only a URL, a key and a model id change between it and NVIDIA's own API.

One runtime, two dependencies, no server to install.

LAYER	WHAT	WHY
Language and runtime	TypeScript on Node 22.19 or later; `node:sqlite` for the ledger	No database server, nothing beyond `npm install`
Runtime dependencies	`yaml` for policy files, `@resvg/resvg-js` to render charts	The only two packages the product needs at run time
Model	NVIDIA Nemotron through an OpenAI-compatible API; recorded replies for offline	The Power rule; tool calling verified
Organisational data	Zetaris in local Docker over three synthetic silos	One statement over file, SQL and REST
Memory and history	Meterless World Model, H-MEM, Markovian with their own conformance suites	Used as written; the suites run against our copies
Signing and proof	Ed25519 from `node:crypto`; a one-file verifier on plain Node	A stranger checks a dossier with the network off
Interface	Fourteen screens on a local read-only server; light and dark; stacked cards on a phone	The UI is the product
Verification	vitest, a pre-registered benchmark, `desk prove`	Each claim runs; NOT RUN is never a pass

No GPU, no training, no package installed at run time, no download at run time. Cursor became optional on 22 October and was not used; this was written with Claude Code as the assistant.

A regression called, with its interval, a flat control arm, and what it cannot see.

CHANGE AGAINST BASELINE

-40.0

points. 99% interval -56.2 to -23.8, on 52 tasks. Called **Regressed** on day 30.

CONTROL ARM

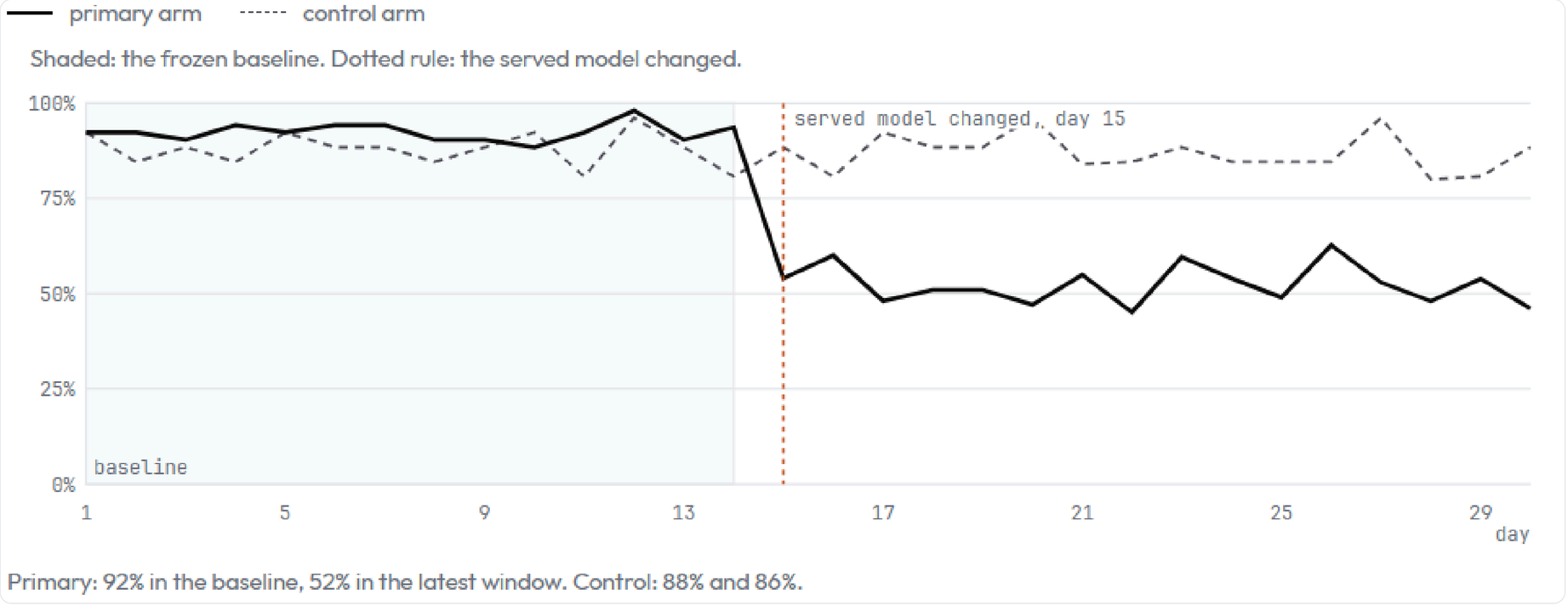
flat

The unchanged model moved -2.2 points (95% interval -6.7 to +2.3).

SMALLEST CHANGE IT CAN SEE

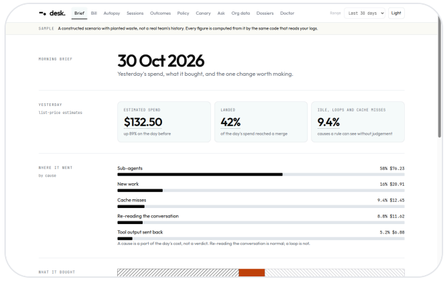
20.7

points in one window. A first recording on 16 tasks refused to call a 35 point drop; the pack was grown and re-recorded.

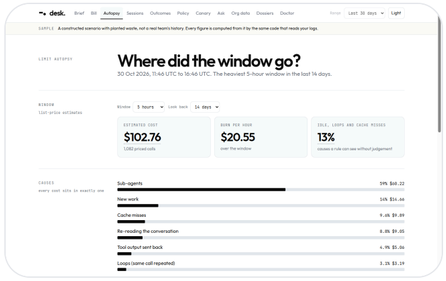


Real model runs, one pack a day on each arm, dates assigned. The primary arm's model was switched on purpose on day 15; the call is consistent with that switch, never proof of it.

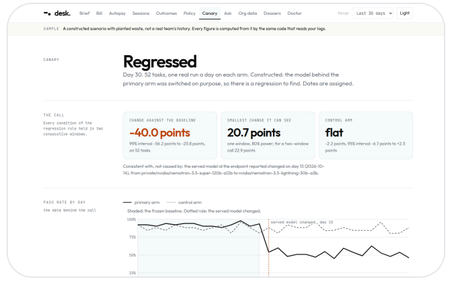
Fourteen screens. Light and dark. Every figure says where it came from.



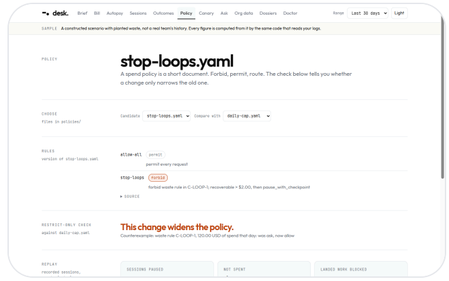
brief



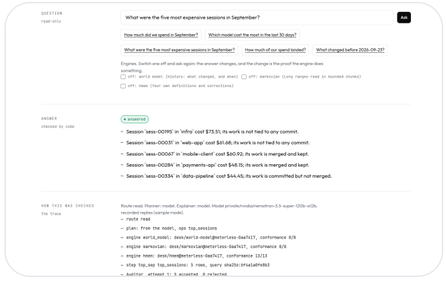
autopsy



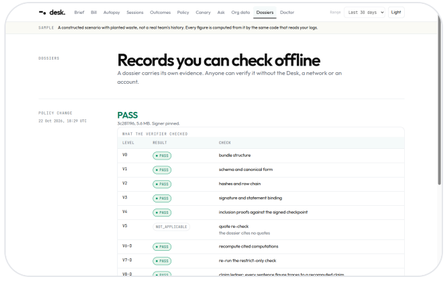
canary



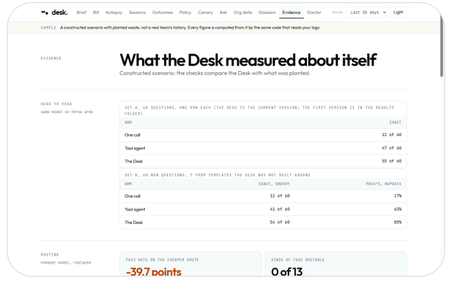
policy



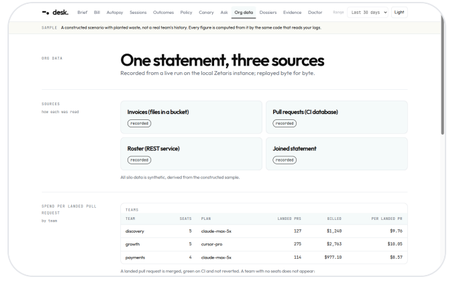
ask



dossiers



evidence



org data

We read their list of reasons to pick a different scope. Here is each one.

A private API key you do not have	None needed for the sample
A dataset you cannot access quickly	Generated from a seed on your machine
Many GB of raw data, no sample strategy	Generated, not shipped; `data/README.md`
GPU training	None
Agents run once in a fixed sequence	Plans are revised after a critic; paths depend on the question
Sponsors with no real role	Each changes the output when off
Value cannot be explained	One sentence, slide 1

Output mostly generic text	Every figure is computed by code and carries its rows
Cannot run without internet	The sample runs offline
Fails when an API is down	Every language role has a rule brain
Manual steps not in the README	`npm run desk -- demo` is the demo
No sample inputs and outputs	`input_examples/`, `output_examples/`
Cannot show how the agent got there	A JSONL trace for each Ask and Propose run

THE SAMPLE CARD

~20 s

offline, from a fresh clone.

THE WHOLE DEMO

< 1 min

offline. Target was 5 to 10 minutes.

ONE LIVE ASK

8 to 80 s

two model calls; retries, turns and wall clock are capped in code.

Say what was not run, and what failed.

RUN, AND PASSING: 61 OF 61 CHECKS

- Planted waste found in a constructed scenario, with false positives counted
- A regression called on 52 real tasks with a flat control arm
- A pre-registered benchmark, 900 runs replayed from recordings
- Gate invariants through 400 random operations
- Real Claude Code blocked a Bash call (and ran it under an allow policy)
- Zetaris stopped: the output changed
- Clean clone to card in 30 seconds, same figures hash

MEASURED, AND SHORT OF ITS LABEL

- The wind tunnel's 90% range held 73% (72% with no change planted)
- Its chance of blocking landed work: calibration error 0.24
- The Desk is still wrong on about a tenth of questions
- Learning moved 2 of 60: no improvement claimed

NOT BUILT, OR NOT VERIFIED

- A swap whose baseline reproduces its recorded passes (the live run was withheld)
- Cursor and Gemini CLI hooks run live
- How a plan's limit maps to tokens
- Zetaris semantic layer and data steward

From a desk to a gate for every agent.

- **Every tool, not one.** Claude Code and Codex are stopped by their hooks, run live; the gateway door stops any tool that can use another base URL (real endpoint over TLS for chat; stand-ins for the messages and responses routes). The Cursor and Gemini CLI hooks pass their contract tests and still need a live run.
- **A swap that can answer.** The replay of your own tasks on a cheaper model is built and its first live run was withheld, because the larger model reproduced only 57% of the recorded passes. It needs tasks of your own with tests that fail before the work and pass after it.
- **A better range.** A one-month range that holds as often as it says: 73% against 90% in 64 constructed months, and a chance of blocking landed work that is too confident.
- **Someone to receive the index.** The opt-in regression index previews its exact bytes and sends only what you accept; nothing yet listens at the other end.
- **Open formats, published.** The ledger and the dossier have an adapter kit and a Sigstore bundle; a standard other desks write is not yet published.

desk.

Run it on your own logs.

```
npm run desk
```

Under three minutes. Entirely local. Your own limit autopsy, your own ranked fixes, and what it could not know.

No logs? npm run desk -- --sample shows the same card offline.